



WhatsApp Bot – Complete Development Requirement

Version: V1 • Initial Production Version

Business Networking • B2B & B2C Commerce • Technology Solutions

Requirement Specification Document

1. Project Overview

Build a WhatsApp bot for GloAro to handle customer enquiries for **GloAro Mart** and **GloAro Digital Network**. The bot provides menu-based conversations, lead capture, PostgreSQL storage, Google Sheets integration and admin WhatsApp notifications.

The architecture should be scalable for future AI, CRM, campaigns, GloAro Tech and order tracking.

2. Initial Scope

Included in V1

- WhatsApp Business API integration with a dedicated company WhatsApp number
- Welcome message and main menu
- GloAro Mart and GloAro Digital Network conversation flows
- Lead capture, PostgreSQL storage and Google Sheets sync
- Admin WhatsApp notification
- Basic admin dashboard with customer & conversation history
- Bot sessions, error handling, testing and production deployment

Deferred (Future Phases)

- Advanced AI conversations
- Marketing campaigns
- Complex CRM and billing
- Multi-tenant SaaS

3. Main Welcome Flow

Customer sends WhatsApp message → Welcome + Main Menu → GloAro Mart or GloAro Digital Network

Welcome Message

👋 Welcome to GloAro!
We're delighted to have you here.

GloAro is your one-stop platform for Business Networking, B2B & B2C Commerce, Technology Solutions, and Business Growth.

Please choose:

- 1 GloAro Mart – 🛒 Buy • Sell • Grow
- 2 GloAro Digital Network – ❤️ Business Networking Community

Reply with the option number.

4. GloAro Mart Flow

Menu Options

1. B2B Marketplace
2. B2C Marketplace
3. Become a Seller
4. Track My Order
5. Contact Sales Team

Option	Content / Details Collected
B2B Marketplace	Categories: Food & FMCG, Electronics, Fashion & Garments, Industrial Products, Other Products. Collect: Company Name, Contact Person, Mobile Number, Product Requirement.
B2C Marketplace	Browse Products, Latest Offers, Customer Support.
Become a Seller	Collect: Business Name, Owner Name, Mobile Number, Product Category, Business Location.
Track My Order	Ask for Order ID. Connect to order API if available; otherwise create a support request.
Contact Sales Team	Collect: Name, Mobile Number, Requirement. Create a lead.

5. GloAro Digital Network Flow

Menu Options

1. About GloAro
2. Membership Benefits
3. Chapters
4. Franchise Opportunity
5. Join GloAro
6. Upcoming Events
7. Contact Team

Option	Content / Details Collected
About GloAro	Explain referrals, connections, expert learning, collaboration and business growth.
Membership Benefits	Explain benefits, then collect: Full Name, Mobile Number, Email, Business Name, Business Category, City.
Chapters	Collect City or PIN Code.
Franchise Opportunity	Explain benefits, then collect: Name, Mobile Number, City.

Join GloAro	Collect: Full Name, Mobile Number, Email, Business Name, Business Category, City.
Upcoming Events	Show Business Meetups, Guest Speaker Sessions, Referral Meetings, Business Growth Workshops. REGISTER collects: Name, Mobile, Email, City.
Contact Team	Show GloAro phone/email and collect requirement.

6. Lead Capture

Customer → Bot collects details → Validate → Create Lead → PostgreSQL → Google Sheets → Admin WhatsApp Notification

Lead Fields

Field	Field	Field
Lead ID	Date & Time	WhatsApp Number
Name	Email	Business Name
Main Category	Subcategory	Product / Business Category
City	Requirement	Status

Status values: New / Follow-up / Closed

7. Admin WhatsApp Notification

 New WhatsApp Bot Lead

Name: [Name]

Phone: [WhatsApp Number]

Category: [Main Category]

Subcategory: [Subcategory]

Requirement: [Requirement]

Time: [Date & Time]

Status: New

8. Google Sheet Integration

Columns: Date & Time, WhatsApp Number, Name, Email, Business Name, Main Category, Subcategory, Product Category, City, Requirement, Status.

Use the Google Sheets API to append each lead, with proper error handling and retries.

9. WhatsApp Integration

Use the official **Meta WhatsApp Business Platform / Cloud API**.

GloAro Provides

Developer Handles

Dedicated business mobile number, Meta Business account, WhatsApp Business account, required access & verification.

Meta app configuration, API setup, credentials, webhook, message handling, bot integration and testing.

One dedicated GloAro business number is used for V1. API tokens must never be exposed in frontend code.

10. Webhook

- **GET** /webhooks/whatsapp — verification
- **POST** /webhooks/whatsapp — message events

Customer → WhatsApp → Meta → Webhook → NestJS Backend → Bot Engine → Response

Webhook events must be logged and processed **idempotently** to prevent duplicate messages and duplicate leads.

11. Admin Panel

Screen	Details
Login	Authenticated admin access.
Dashboard	Total Leads, New Leads, Follow-up, Closed, Today's Leads, Customers, Active Conversations.
Leads	View, Search, Filter, Status, Notes, Customer Number, Requirement.
Customers	Name, WhatsApp Number, Business, Category, Last Interaction, Conversation History.
Conversations	Customer, Messages, Date/Time, Status, Agent.
Bot Menu	Welcome message, main menu, submenus, responses, images and links.
Settings	Company details, WhatsApp config, admin notification number, Google Sheet.

12. Database

PostgreSQL. Initial tables:

Table	Table	Table
users	customers	leads
conversations	messages	bot_flows
bot_sessions	whatsapp_accounts	settings

Relationships: Customer → Conversations → Messages | Customer → Leads

13. Recommended Technology

Layer	Stack
Frontend	React.js, Vite, Tailwind CSS, React Query, Axios
Backend	NestJS, Prisma, REST API
Database	PostgreSQL
WhatsApp	Meta WhatsApp Business Platform
Google	Google Sheets API
Deployment	AWS, Nginx, HTTPS

14. Development Phases

Phase	Name	Deliverables
1	Requirement Finalization	Messages, menus, lead fields, notification, Sheet, business contacts.
2	UI / UX	Login, Dashboard, Leads, Customers, Conversations, Bot Menu, Settings.
3	Database	PostgreSQL, Prisma, tables and relationships.
4	Backend	Authentication, Customer, Lead, Conversation, Message, Bot and Settings APIs.
5	Bot Engine	Welcome, Main Menu, GloAro Mart and Digital Network flows.
6	Lead Capture	Questions, validation, lead creation and status.
7	Google Sheets	Connect and append leads.
8	Admin Notification	Send new lead notification to admin WhatsApp.
9	Admin Panel	Connect frontend and backend.
10	WhatsApp API	Meta Business, WhatsApp account, company number, API, webhook, send/receive.
11	Testing	All menus, invalid input, interrupted flow, duplicate messages, webhook, Google Sheet and API failures.
12	Production	Deployment, domain, HTTPS, environment variables, credentials, backups, logs and monitoring.

15. What GloAro Must Provide Initially

Category	Items
Business	GloAro logo, company name, company description, official email, official website, customer support number.
WhatsApp	Dedicated GloAro mobile number, Meta Business account, WhatsApp Business account, required Meta access/verification.
Content	Final approved Welcome message, GloAro Mart content, Digital Network content, contact information, product/service information, event information.
Lead Management	Admin notification WhatsApp number, sales contact, membership contact, franchise contact, Google account and Google Sheet.
Technical	AWS / server access, domain or subdomain, production email if required.

16. What the Developer Will Handle

Frontend, backend, database, bot engine, WhatsApp API, webhook, Google Sheets API, lead capture, admin notification, admin dashboard, authentication, validation, error handling, testing and deployment.

17. Final Architecture

Customer → WhatsApp → Meta WhatsApp Platform → Webhook → NestJS Backend → Bot Engine → PostgreSQL

- Bot Engine branches into **GloAro Mart** and **GloAro Digital Network**.
- Leads flow to **Google Sheets** and **Admin WhatsApp**.
- **React Admin Panel** connects to the backend.

18. V1 Completion Criteria

- Customer can message the GloAro WhatsApp number and the welcome message works
- All GloAro Mart and Digital Network flows work end to end
- Required details are collected correctly
- Leads are saved to PostgreSQL and Google Sheets
- Admin receives the WhatsApp notification
- Admin can view and update leads
- Conversation history is stored
- Invalid inputs are handled gracefully
- Production WhatsApp works reliably
- Webhook is secured and tested

19. Immediate Starting Requirements

Before Development Begins

1. Dedicated WhatsApp mobile number
2. Meta Business / WhatsApp Business access
3. Final approved conversation content
4. Admin notification number
5. Google account + Google Sheet
6. GloAro business and contact information

Development Order

Requirement → Conversation Flow → Database → Backend → Bot → Lead Capture → Google Sheets → Admin Notification → WhatsApp API → Testing → Production